

Food Analysis & Analytical 2200) For 2nd year
AHuN students

Badasa Rata
Department of Human Nutrition
Faculty of Chemical & Food Engineering
Bahirdar Institute of Technology

Sept, 2021

1. Introduction to food analysis (4hrs)

- Food: is any substance consumed to provide nutritional support for the body.
- Food analysis: is the discipline dealing with the development, application and study of analytical procedures for characterizing the **properties of foods and their constituents**
- Food analysis also refers to the assessment of food composition using **chemical, biological or physical methods**.
- The analytical procedures are used to provide information about a wide variety of different characteristics of foods, including their composition, structure, physicochemical properties and sensory attributes.

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1.1 Chemical composition of food

- Food contains essential nutrients, such as carbohydrates, fats, proteins, vitamins, or minerals.

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Food composition

- In order to assess the quality and safety of food products



- Chemical and physical properties of food need to be known



Analysis of the chemical composition

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Food composition

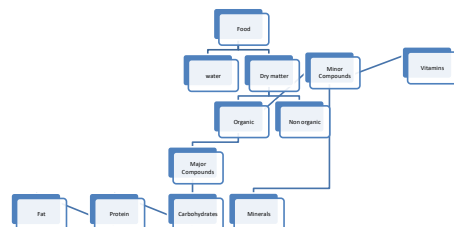
- Quality of food: nutritional, sensorial and functional quality
- Safety of a food: microbial and chemical

Acceptability of food product

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Food composition



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1.2. Methods of food analysis

- These are
 - qualitative,
 - quantitative and
 - instrumental methods of food analysis

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Qualitative methods

- A qualitative analysis determines the presence or absence of a particular compound, but not the mass or concentration.

Quantitative methods

- Quantitative Analysis refers to analyses in which the amount or concentration of an analyze may be determined (estimated) and expressed as a numerical value in appropriate units

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- Volumetric (as the name suggests) involves measuring the **relative volumes of two solutions**, one of known concentration, that react together. Usually this would be done by performing a titration.
- Gravimetric analysis involves the **weighing of the unknown** before and after carrying out various procedures on it. For instance you might thermally decompose it; or you might dissolve it, and then precipitate and eventually weigh, a solid from that solution.

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- Gravimetric analysis yields **more accurate data** about the composition of a sample than volumetric analysis does, but the first one takes more time to perform in the laboratory.
- Volumetric analysis in the other side doesn't take that much time and the results that we obtain are in the most cases satisfactory.
- Volumetric analysis can be simply a titration based in a neutralization reaction but it can also be a precipitation or a complex forming reaction as well as a **titration based in a redox reaction**

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Instrumental analysis

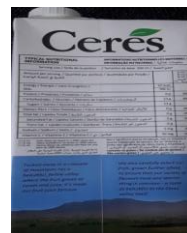
- **Instrumental analysis** is a field of analytical chemistry that investigates analytes using scientific instruments.
- E.g. spectroscopy, chromatography, HPLC etc

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Purposes of Food Analysis

- **Government Regulations and Recommendations:** USDA, FDA, EFDA
- **Mandatory Standards Eg. Composition of peanut**



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- **Nutritional Labeling**
 - FDA claims for Ca - osteoporosis, Na-high blood pressure, cholesterol and soluble fiber-heart disease
- **Authenticity:** highly valued
Eg. premium coffee may claim that the coffee beans are from Ethiopia
- **Research and Development:**
 - the development of new products,
 - the improvement of existing products and the reduction of manufacturing costs

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- **Food Inspection and Grading:**
 - to ensure that products meet the appropriate laws and regulations
- **Food Safety**
 - **harmful microorganisms** (*e.g.*, Listeria, Salmonella),
 - **toxic chemicals** (*e.g.*, pesticides, herbicides) or
 - **extraneous matter** (*e.g.*, glass, wood, metal, insect matter).
- **Quality control:** Food processing company analyzes raw material, on process and end product
- How can food manufacturers control variations ?
 - Characterization of raw materials
 - Monitoring of food properties during processing
 - Characterization of final product

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Choosing an Analytical Techniques

- The analytical technique selected depends on the property to be **measured**, **the type of food to be analyzed**, and **the reason for carrying out the analysis**.
- Some of the criteria that are important in selecting an appropriate technique to analyse foods are listed below:
- **Precision:** A measure of the ability to reproduce an answer between determinations performed by the same scientist (or group of scientists) using the same equipment and experimental approach.

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- **Reproducibility:** A measure of the ability to reproduce an answer by scientists using the same experimental approach but in different laboratories using different equipment
- **Accuracy:** A measure of how close one can actually measure the *true value* of the parameter being measured, *e.g.*, fat content, or sodium concentration
- **Simplicity of operation:** A measure of the ease with which relatively unskilled workers may carry out the analysis.

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- **Cost:** The total cost of the analysis, including the reagents, instrumentation and salary of personnel required to carry it out
- **Speed:** The time needed to complete the analysis of a single sample or the number of samples that can be analyzed in a given time
- **Sensitivity:** A measure of the lowest concentration of a component that can be detected by a given procedure

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- **Specificity:** A measure of the ability to detect and quantify specific components within a food material, even in the presence of other similar components, *e.g.*, fructose in the presence of sucrose or glucose
- **Safety:** Many reagents and procedures used in food analysis are potentially hazardous *e.g.* strong acids or bases, toxic chemicals or flammable materials.
- **Destructive/Non destructive:** In some analytical methods the sample is destroyed during the analysis, whereas in others it remains intact.

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1.3 Sampling and sample preparation

- The small portions taken for analysis are referred to as **samples**, and the **entire lot or the entire production** for a certain period of time, in the case of continuous processes, is called a **population**. The **process of taking** samples from a population is called **sampling**.

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1.3.1. SELECTION OF SAMPLING PROCEDURES

- defining the population that is going to be sampled
- Once sampling is conducted, a series of stepwise procedures – from sample preparation, laboratory analysis, data processing, and interpretation – is needed to obtain data from the samples
- The **International Union of Pure and Applied Chemistry (IUPAC)** defines a **sampling plan** as: “A predetermined procedure for the selection, withdrawal, preservation, transportation, and preparation of the portions to be removed from a lot as samples”

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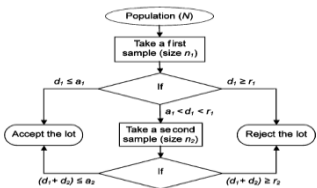
1.3.2 Sampling Plan

- The most important use of sampling are
 1. Nutritional labeling
 2. Detection of contaminants and foreign matter
 3. Statistical process control (Quality Assurance)
 4. Acceptance of raw materials, ingredients, or products (Acceptance Sampling)
 5. Release of lots of finished product
 6. Detection of adulterations
 7. Microbiological safety
 8. Authenticity of food ingredients, etc.

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1.3.5 Acceptance Sampling



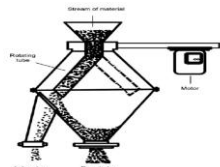
5-1 figures Example of a double sampling plan with two points where the decision of acceptance or rejection can be made [Adapted from (7)]. N , population size; n_1 and n_2 , sample size; a_1 and a_2 , acceptance numbers; r_1 and r_2 , rejection numbers; d_1 and d_2 , number of nonconformities. Subindices 1 and 2 represent samples 1 and 2, respectively.

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1.4 PREPARATION OF SAMPLES

- To obtain a smaller quantity for analysis the sample can be spread on a clean surface and divided into quarters. The two opposite quarters are combined.



5-4 A rotating tumbler divider for reducing a large sample to a smaller, representative portion. (Courtesy of Glen Mills, Inc., Chilton, NJ.)

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1.3.3 Factors Affecting the Choice of Sampling Plans

Factors to Be Considered	Questions
Purpose of the inspection	Is it to accept or reject the lot? Is it to measure the average quality of the lot? Is it to determine the variability of the product?
Nature of the product	Is it homogeneous or heterogeneous? What is the unit size? How consistently have past populations met specifications? What is the cost of the material being sampled?
Nature of the test method	Is the test critical or minor? Will someone become sick or die if the population fails to pass the test? Is the test destructive or nondestructive? How much does the test cost to complete?
Nature of the population being investigated	Is the lot large but uniform? Does the lot consist of smaller, easily identifiable sublots? What is the distribution of the units within the population?

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Assignment (Individual)

- 1. Assume you are assigned as a laboratory technician at Bless Agri food lab. Describe the detail activities you perform on sampling and sample preparation techniques for agriculture products from retailers, wholesalers, producers and farmers. The food matrices include dairy and dairy products, fruits and fruit products, confectionery, meat and meat products, cereal and cereal products.

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2. PRINCIPLES AND APPLICATIONS OF ANALYTICAL TECHNIQUES

- Gravimetric methods** are quantitative methods that are based on determining the mass of a pure compound to which the analyte is chemically related
- The mass measurement is made with an analytical balance, an instrument that yields highly accurate and precise data

Types of Gravimetric Methods

- There are three types of gravimetric methods of analysis which are: **precipitation gravimetry**, **volatilization gravimetry**, **electrogravimetry**
- In precipitation gravimetry, the analyte is separated from a solution of the sample as a **precipitate** and is converted to a compound of known composition that can be weighed.
- In volatilization gravimetry, the analyte is separated from other constituents of a sample by conversion to a gas of known chemical composition.
- In electrogravimetry, the analyte is separated by deposition on an electrode by an electrical current

Titrimetric Methods

- Titration is a quantitative method of analysis in which an equivalent quantity of standard solution (titrant) is added systematically from burette to a known volume of sample (analyte)
- It is performed by slowly adding a standard solution from a buret or other liquid-dispensing device to a solution of the analyte until the reaction between the two is judged complete.
- The volume or mass of reagent needed to complete the titration is determined from the difference between the initial and final readings.

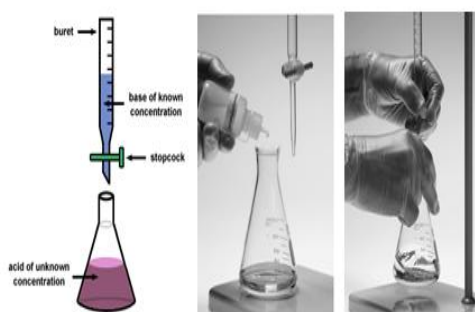


Figure 2.1 Schematic diagrams for titration apparatus

Terms Used in Volumetric Titrimetry

- Titration:** a process in which a standard reagent is added to a solution of an analyte until the reaction between the analyte and reagent is judged to be complete
- The equivalence point** in a titration is a theoretical point reached when the amount of added titrant is chemically equivalent to the amount of analyte in the sample.
- The end point** is the point in a titration when a physical change occurs that is associated with the condition of chemical equivalence
- Titration error:** The difference in volume or mass between the equivalence point and the end point.
- A standard solution (or a standard titrant):** a reagent of known concentration that is used to carry out a titrimetric analysis

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- The ideal standard solution for a titrimetric method will
 - be sufficiently stable so that it is necessary to determine its concentration only once;
 - react rapidly with the analyte so that the time required between additions of reagent is minimized;
 - react more or less completely with the analyte so that satisfactory end points are realized; and
 - undergo a selective reaction with the analyte that can be described by a balanced equation.

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- **Primary Standards:** A primary standard is a **highly purified compound** that serves as a reference material in volumetric and mass titrimetric methods.

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- Important requirements for a primary standard are the following:
 - High purity. Established methods for confirming purity should be available.
 - Atmospheric stability
 - Absence of hydrate water so that the composition of the solid does not change with variations in humidity.
 - Modest cost
 - Reasonable solubility in the titration medium.
 - Reasonably large molar mass so that the relative error associated with weighing the standard is minimized

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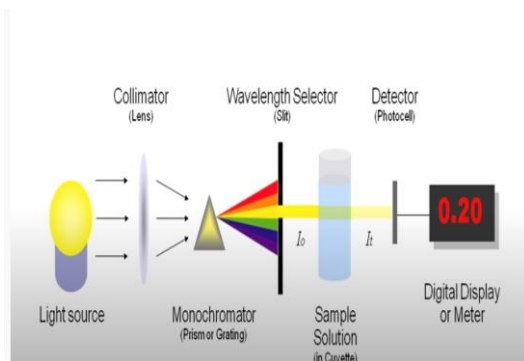
- Standardization is the process of making standard solutions
- In standardization, the concentration of a volumetric solution is determined by **titrating it against a carefully measured** quantity of a primary or secondary standard or an exactly known volume of another standard solution.

Types of titrimetry

- **Gravimetric titrimetry:** determines the quantity of analyte by a measure of the **mass** of a solution of known concentration.
- **Volumetric titrimetry:** involves measuring the **volume** of a solution of known concentration that is needed to react essentially completely with the analyte.
- The volume of titrant required to just completely react with the analyte is the ***Titre***.
- **Precipitation Titrimetry** is a type of volumetric titration where the end product of the analyte and the titrant's reaction is a slightly soluble precipitate/solid
- **Coulometric titrimetry:** is a type of titrimetry in which the quantity of **charge in coulombs** required to complete a reaction with the analyte is the measured quantity.

Colorimetric methods

- These methods rely on a **change in color** of a reagent when it reacts with a specific mineral in solution which can be quantified by measuring the absorbance of the solution at a specific wavelength using a spectrophotometer.
- Colorimetric methods are used to determine the concentration of a wide variety of different minerals
- Eg. banana



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Spectroscopy

- **Spectroscopy** deals with the production, measurement, and interpretation of spectra arising from the interaction of electromagnetic radiation with matter
- Spectral measurement devices are referred to as spectrometers, spectrophotometers, spectrographs or spectral analyzers
- Spectroscopic methods based on the absorption or emission of radiation in the ultraviolet (UV), visible (Vis), infrared (IR), and radio, nuclear magnetic resonance,

General methods of spectroscopy

Terms In spectroscopy

- **Absorption:** is the process by which energy is transferred from an electromagnetic wave to an atom or molecule and causes it to move to an excited state.
- **Absorption** can only occur when an atom or molecule absorbs a photon of light that has an energy which exactly corresponds to the difference between two energy levels,
- **Emission:** The sample being analyzed is energetically stimulated (*e.g.*, by heating or application of radiation) and the amount of electromagnetic radiation produced by the sample is measured at different wavelengths, *e.g.*, atomic emission spectroscopy, fluorescence.

Cont...

- **Transmission:** An electromagnetic wave generated by the analytical instrument is **propagated directly through the sample and the reduction in its amplitude** due to interaction with the sample is measured at different wavelengths, *e.g.*, AAS, infrared transmission measurements, UV-visible spectrophotometry.
- **Reflection:** An electromagnetic wave generated by the analytical instrument is **reflected from the surface** of the sample and the reduction in its amplitude due to interaction with the sample is measured at different wavelengths, *e.g.*, infrared reflection measurements, color measurements

Atomic Spectroscopy

- The determination of **mineral type** and **concentration** by atomic spectroscopy is more sensitive, specific, and quicker than traditional wet chemistry methods.

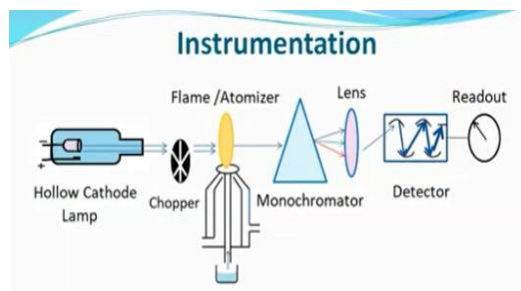
Principles of Atomic Spectroscopy

- The primary cause of absorption and emission of radiation in atomic spectroscopy is *electronic transitions* of outer shell electrons
- The samples used in atomic spectroscopy are individual atoms in a gaseous state, whereas those used in UV-visible spectroscopy are molecules dissolved in liquids.
- In atomic spectroscopy the peaks are narrow and well defined, but in UV-visible spectroscopy they are broad and overlap with one another

Reasons why AS peak is narrow and well defined

- Firstly, because absorption or emission is from atoms, rather than molecules, there are no vibrational or rotational transitions superimposed on the electronic transitions
- Secondly, because the atoms are in a gaseous state they are well separated from each other and do not interact with neighbouring molecules

Summary of AAS



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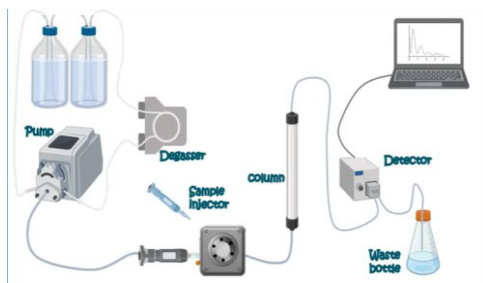
Chromatography

- It is technique that enables the separation, identification and purification of components of mixture for qualitative and quantitative analysis
- The chromatography system consist of a **stationary phase** which can be a solid or a liquid supported on a solid; and a **mobile phase** which flows continuously around the stationary phase.
- The mobile phase can be gaseous or liquid. Depending on the nature of the mobile and stationary phases, the following types of chromatography can be achieved:
 - Gas-Solid (GS),
 - Gas-Liquid (GL),
 - Liquid- Liquid (LL) and
 - Liquid-Solid (LS) chromatography.

Chromatography terms

- The **analyte** is the substance to be separated during chromatography.
- **Analytical chromatography** is used to determine the existence and possibly also the concentration of analyte(s) in a sample.
- A **bonded phase** is a stationary phase that is covalently bonded to the support particles or to the inside wall of the column tubing.
- A **chromatogram** is the visual output of the chromatograph.
- In the case of an optimal separation, different peaks or patterns on the chromatogram correspond to different components of the separated mixture.

HPLC diagram



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- **The solute** refers to the sample components in partition chromatography.
- **The solvent** refers to any substance capable of solubilizing another substance, and especially the liquid mobile phase in liquid chromatography.
- **The stationary phase** is the substance fixed in place for the chromatography procedure. Examples include the silica layer in thin layer chromatography
- **The detector** refers to the instrument used for qualitative and quantitative detection of analytes after separation.

3. ANALYSIS OF FOOD COMPONENTS

- **Proximate analysis** of foods refers to determining the major components of foods like moisture, ash (total minerals), lipids, protein, and carbohydrates

Determination of Moisture and Total Solids

- It is important to food scientists for a number of different reasons:
 - Legal and Labeling Requirements:
 - Economic:
 - Microbial Stability:
 - Food Quality:
 - Food Processing Operations: *e.g.* mixing, drying, flow through a pipe or packaging.

Types of water

- **Bulk water:** Bulk water is free from any other constituents, so that each water molecule is surrounded only by other water molecules.
- **Capillary or trapped water:** Capillary water is held in narrow channels between certain food components because of capillary forces
- **Physically bound water:** A significant fraction of the water molecules in many foods are not completely surrounded by other water molecules, but are in molecular contact with other food constituents, *e.g.* proteins, carbohydrates or minerals
- **Chemically bound water:** Some of the water molecules present in a food may be chemically bonded to other molecules as water of crystallization or as hydrates, *e.g.* $\text{NaSO}_4 \cdot 10\text{H}_2\text{O}$. These bonds are much stronger than the normal water-water bond

Methods of Moisture Analysis

1. Evaporation Methods (Oven drying method) Evaporation Devices (Instruments used)

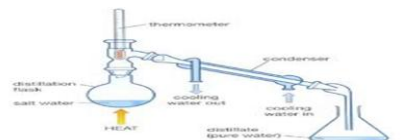
- Convection and forced draft ovens, Vacuum oven, Microwave oven, Infrared lamp drying:

2. Distillation Methods

- Distillation methods are based on *direct* measurement of the amount of water removed from a food sample by evaporation: % Moisture = $(M_{\text{WATER}}/M_{\text{INITIAL}}) \times 100$
- In contrast, evaporation methods are based on *indirect* measurement of the amount of water removed from a food sample by evaporation:
- %Moisture = $((M_{\text{INITIAL}} - M_{\text{DRIED}})/M_{\text{INITIAL}}) \times 100$

Dean and Stark Method

- A known weight of food is placed in a flask with an organic solvent such as xylene or toluene
- The organic solvent must be insoluble with water; have a higher boiling point than water; be less dense than water; and be safe to use
- The flask containing the sample and the organic solvent is attached to a condenser by a side arm and the mixture is heated.
- The water in the sample evaporates and moves up into the condenser where it is cooled and converted back into liquid water,



Advantages and Disadvantages

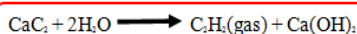
- Advantages: Suitable for application to foods with low moisture contents; Suitable for application to foods containing volatile oils, such as herbs or spices,
- Disadvantages: Destructive; Relatively time-consuming; Involves the use of flammable solvents; Not applicable to some types of foods

3. Chemical Reaction Methods

- Reactions between water and certain chemical reagents can be used as a basis for determining the concentration of moisture in foods.
- In these methods a chemical reagent is added to the food that reacts specifically with water to produce a measurable change in the properties of the system, *e.g.*, mass, volume, pressure, pH, color, conductivity
- To make accurate measurements it is important that the chemical reagent reacts with all of the water molecules present, but not with any of the other components in the food matrix.
- Two methods that are commonly used in the food industry are the
 - A. Karl-Fisher titration and (reading assignment)
 - B. Gas production methods

B. Gas production methods

- Commercial instruments are also available that utilize specific reactions between chemical reagents and water that lead to the production of a gas.
- For example, when a food sample is mixed with powdered calcium carbide the amount of acetylene gas produced is related to the moisture content.
- The amount of gas produced can be measured in a number of different ways, including (i) the volume of gas produced, (ii) the decrease in the mass of the sample after the gas is released, and (iii) the increase in pressure of a closed vessel containing the reactants



Analysis of Ash

- The **'ash content'** is a measure of the total amount of minerals present within a food, whereas the **'mineral content'** is a measure of the amount of specific inorganic components present within a food, such as Ca, Na, K, Cl and so on
- Determination of the ash and mineral content of foods is important for a number of reasons:
- Nutritional labelling: The concentration and type of minerals present must often be stipulated on the label of a food.
- Microbiological stability: High mineral contents are sometimes used to retard the growth of certain microorganisms
- Quality: The quality of many foods depends on the concentration and type of minerals they contain, including their taste, appearance, texture and stability.

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- Although most minerals have fairly low volatility at these high temperatures, some are volatile and may be partially lost, *e.g.*, iron, lead and mercury.
- If an analysis is being carried out to determine the concentration of one of these substances then it is advisable to use an alternative ashing method that uses lower temperatures.
- The ash content can be expressed on either a *dry* or *wet* basis

Calculate the percentage ash content (wet weight basis) as follows:

$$\% \text{ ash (wet)} = \frac{(\text{Wt. crucible \& ash} - \text{wt. crucible})}{(\text{Weight before ashing (wt. crucible \& sample} - \text{wt. crucible})} \times 100$$

Calculation of ash content on dry basis (when moisture content is known) as follows:

$$\% \text{ ash (Dry)} = \frac{\% \text{ ash (wet)}}{(100 - \% \text{ moisture})} \times 100$$

4. Physical Methods

- A number of analytical methods have been developed to determine the moisture content of foods that are based on the fact that water has appreciably different bulk physical characteristics than the food matrix, *e.g.* density, electrical conductivity or refractive index.
- These methods are usually only suitable for analysis of foods in which the composition of the **food matrix does not change significantly**, but the ratio of water-to-food matrix changes
- For example, the water content of oil-in-water emulsions can be determined by measuring their density or electrical conductivity because the density and electrical conductivity of water are significantly higher than those of oil.

Methods of Ash Content Determination

1. Dry Ashing

- Dry ashing use a high temperature **muffle furnace** capable of maintaining temperatures of between 500 and 600 °C.
- Water and other volatile materials are vaporized and organic substances are burned in the presence of the oxygen in air to CO₂, H₂O and N₂.
- Most minerals are converted to oxides, sulfates, phosphates, chlorides or silicates.



Figure 3.4 A: Muffle furnace B: different size Crucibles

2. Wet Ashing

- Wet ashing is primarily used in the preparation of samples for subsequent analysis of specific minerals.
- It breaks down and removes the organic matrix surrounding the minerals so that they are left in an aqueous solution
- A dried ground food sample is usually weighed into a flask containing strong acids and oxidizing agents (*e.g.*, nitric, perchloric and/or sulphuric acids) and then heated.

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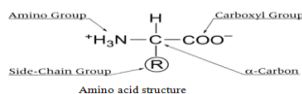
- Heating is continued until the organic matter is completely digested, leaving only the mineral oxides in solution
- The temperature and time used depends on the type of acids and oxidizing agents used. Typically, a digestion takes from 10 minutes to a few hours at temperatures of about 350°C. The resulting solution can then be analyzed for specific minerals.
- Advantages: Little loss of volatile minerals occurs because of the lower temperatures used, more rapid than dry ashing.
- Disadvantages: Labor intensive, requires a special fume-cupboard if perchloric acid is used because of its hazardous nature, low sample throughput.

Determination of Water Soluble and Insoluble Ash

- As well as the total ash content, it is sometimes useful to determine the ratio of water soluble to water-insoluble ash as this gives a useful indication of the quality of certain foods, *e.g.*, the fruit content of preserves and jellies.
- Ash is diluted with distilled water then heated to nearly boiling, and the resulting solution is filtered.
- The amount of soluble ash is determined by drying the filtrate, and the insoluble ash is determined by rinsing, drying and ashing the filter paper.

Analysis of Proteins

- Proteins are polymers of amino acids.
- They are composed of elements including hydrogen, carbon, nitrogen, oxygen, and sulphur
- Twenty α -amino acids are the building blocks of proteins; the amino acid residues in a protein are linked by peptide bonds



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- Proteins are important constituents of foods for a number of different reasons.
- They are a major source of *energy*, as well as containing essential amino-acids, such as lysine, tryptophan, methionine, leucine, isoleucine and valine, which are essential to human health, but which the body cannot synthesize.
- Proteins are also the major structural components of many natural foods, often determining their overall texture, *e.g.*, tenderness of meat or fish products.
- Isolated proteins are often used in foods as ingredients because of their unique functional properties, *i.e.*, their ability to provide desirable appearance, texture or stability
- Typically, proteins are used as gelling agents, emulsifiers, foaming agents and thickeners

Total Protein Concentration Determination

1. Kjeldahl method

- A food is digested with a strong acid so that it releases nitrogen which can be determined by a suitable titration technique.
- The amount of protein present is then calculated from the nitrogen concentration of the food
- The Kjeldahl method can conveniently be divided into three steps: digestion, neutralization and titration

Step 1: Digestion

- The food sample to be analyzed is weighed into a *digestion flask* and then digested by heating it in the presence of sulphuric acid (an oxidizing agent which digests the food), anhydrous sodium sulphate (to speed up the reaction by raising the boiling point) and a catalyst, such as copper, selenium, titanium, or mercury (to speed up the reaction)
- Digestion converts any nitrogen in the food (other than that which is in the form of nitrates or nitrites) into ammonia, and other organic matter to CO_2 and H_2O .
- Ammonia gas is not liberated in an acid solution because the ammonia is in the form of the ammonium ion (NH_4^+) which binds to the sulphate ion (SO_4^{2-}) and thus remains in solution

Step 2:Distillation (Neutralization)

- The solution in the digestion flask is then made alkaline by addition of sodium hydroxide, which converts the ammonium sulfate into ammonia gas:
- $\text{NH}_3 + \text{H}_3\text{BO}_3$ (boric acid) $\rightarrow \text{NH}_4^+ + \text{H}_2\text{BO}_3^-$ (borate ion)
- The ammonia gas that is formed is liberated from the solution and moves out of the digestion flask and into the receiving flask - which contains an excess of boric acid.
- The low pH of the solution in the receiving flask converts the ammonia gas into the ammonium ion, and simultaneously converts the boric acid to the borate ion:

Step 3:Titration

- The nitrogen content is then estimated by titration of the ammonium borate formed with standard sulfuric or hydrochloric acid, using a suitable indicator to determine the end-point of the reaction.
- The concentration of hydrogen ions (in moles) required to reach the end-point is equivalent to the concentration of nitrogen that was in the original food

Calculation

$$\%N = \frac{V\text{ HCl}(L) \times N\text{ HCl} \times 14}{\text{sample wt.}} \times 100$$

$$\% \text{Protein} = F \times \%N$$

Table 3.2 Nitrogen to Protein conversion factors for some Foods

Foods	factors
Egg or meat	6.25
Dairy products	6.38
Wheat	5.70
Other cereal grains and oilseeds	6.25
Almonds	5.18
Peanuts	5.46
Other tree nuts and coconut	5.30

Advantages and Disadvantages

- **Advantages:** The Kjeldahl method is widely used internationally and is still the standard method for comparison against all other methods.
- Its universality, high precision and good reproducibility have made it the major method for the estimation of protein in foods.
- **Disadvantages:** It does not give a measure of the true protein, since all nitrogen in foods is not in the form of protein. Different proteins need different correction factors because they have different amino acid sequences.
- The use of concentrated sulphuric acid at high temperatures poses a considerable hazard, as does the use of some of the possible catalysts. The technique is time consuming to carry-out.

2.Enhanced Dumas method

- automated instrumental technique developed which is capable of rapidly measuring the protein concentration of food samples
- A sample of known mass is combusted in a high temperature (about 900°C) chamber in the presence of O₂. This leads to the release of CO₂, H₂O and N₂
- The CO₂ and H₂O are removed by passing the gasses over special columns that absorb them. The nitrogen content is then measured by passing the remaining gasses through a column that has a thermal conductivity detector at the end
- The column helps separate the nitrogen from any residual CO₂ and H₂O that may have remained in the gas stream

Advantages and Disadvantages

- Advantages: It is much faster than the Kjeldahl method (under 4 minutes per measurement, compared to 1-2 hours for Kjeldahl).
- It doesn't need toxic chemicals or catalysts. Many samples can be measured automatically. It is easy to use.
- Disadvantages: High initial cost. It does not give a measure of the true protein, since all nitrogen in foods is not in the form of protein.

Analysis of Carbohydrates

- Molecules in which the carbohydrates are covalently attached to proteins are known as *glycoproteins*, whereas those in which the carbohydrates are covalently attached to lipids are known as *glycolipids*

$$\% \text{carbohydrates} = 100 - \% \text{moisture} - \% \text{protein} - \% \text{lipid} - \% \text{mineral}$$

Cont...

- Carbohydrates are separated on the basis of their differential adsorption characteristics by passing the solution to be analyzed through a column.
- Carbohydrates can be separated on the basis of **polarities or sizes**, depending on the type of column used.
- HPLC is currently the most important chromatographic method for analyzing carbohydrates because it is capable of rapid, specific, sensitive and precise measurements
- The carbohydrates are then separated on the basis of their size: the smaller the size of a carbohydrate molecule, the faster it moves in an electrical field.

Assignment 2

Describe the principle of the following protein content determination method

1. Direct measurement at 280nm
2. Biuret Method
3. Lowry Method
4. Dye binding methods
5. Turbimetric method
6. Thin layer chromatography (TLC) for carbohydrate content determination

Monosaccharides and Oligosaccharides determination

1. Chromatographic and Electrophoretic methods

- Chromatographic methods are the most powerful analytical techniques for the analysis of the *type* and *concentration* of monosaccharides and oligosaccharides in foods
- Thin layer chromatography (TLC), Gas chromatography (GC) and High Performance Liquid chromatography (HPLC) are commonly used to separate and identify carbohydrates

Cont...

2. Chemical methods

- A number of chemical methods used to determine monosaccharides and oligosaccharides are based on the fact that many of these substances are *reducing agents* that can react with other components to yield precipitates or colored complexes which can be quantified
- The concentration of carbohydrate can be determined gravimetrically, spectrophotometrically or by titration.

Cont...

3. Titration Methods

- The Lane-Eynon method is an example of a titration method of determining the concentration of reducing sugars in a sample
- A burette is used to add the carbohydrate solution being analyzed to a flask containing a known amount of boiling **copper sulphate** solution and a **methylene blue** indicator.
- The reducing sugars in the carbohydrate solution react with the copper sulphate present in the flask.
- Once all the copper sulphate in solution has reacted, any further addition of reducing sugars causes the indicator to change from blue to white.
- The volume of sugar solution required to reach the end point is recorded

Cont...

4. Gravimetric Methods

- **reducing sugar + Cu^{2+} + base $\rightarrow$ oxidized sugar + CuO_2 (precipitate)**
- The **Munson and Walker method** is an example of a gravimetric method of determining the concentration of reducing sugars in a sample.
- Carbohydrates are oxidized in the presence of heat and an excess of **copper sulphate** and **alkaline tartrate** under carefully controlled conditions which leads to the formation of a copper oxide precipitate:
- The amount of precipitate formed is directly related to the concentration of reducing sugars in the initial sample

Cont...

6. Enzymatic Methods

- Analytical methods based on enzymes rely on their **ability to catalyze specific reactions**.
- These methods are rapid, highly specific and sensitive to low concentrations and are therefore ideal for determination of carbohydrates in foods

Cont....

- The disadvantages of this method are
 - the results depend on the precise reaction times, temperatures and reagent concentrations used
 - it cannot distinguish between different types of reducing sugar, and
 - it cannot directly determine the concentration of non-reducing sugars,
 - it is susceptible to interference from other types of molecules that act as reducing agents

Cont...

5. Colorimetric Methods

- The **Anthrone method** is an example of a colorimetric method of determining the concentration of the total sugars in a sample.
- Sugars react with the anthrone reagent under acidic conditions to yield a **blue-green** color
- The sample is mixed with sulphuric acid and the anthrone reagent and then boiled until the reaction is completed. The solution is then allowed to cool and its absorbance is measured at 620 nm. There is a linear relationship between the absorbance and the amount of sugar that was present in the original sample.
- **Dis Adv:** This method determines both reducing and non-reducing sugars because of the presence of the strongly oxidizing sulphuric acid.

7. Density

- The density of a material is its mass divided by its volume. The density of aqueous solutions increases as the carbohydrate concentration increases.
- Thus the carbohydrate concentration can be determined by measuring density, *e.g.*, using density bottles or hydrometers.
- This technique is routinely used in industrial determination of carbohydrate concentration of juices and beverages.



Figure 3.15 Hydrometer

Cont...

8. Immunoassays

- **Immunoassays specific** for low molecular weight carbohydrates are developed by attaching the carbohydrate of interest to a protein, and then injecting it into an animal. With time the animal develops antibodies specific for the carbohydrate molecule.

Analysis of Polysaccharides and Fiber

- Analysis of Fibers
- Dietary fiber is plant polysaccharides that are indigestible by humans, plus lignin.
- **Lignin** is a constituent of the cell walls of almost all dry land plant cell walls.
- Major Components of Dietary Fiber :cellulose, hemicellulose, pectin, and lignin.
- Lignin is usually associated with cellulose and hemicelluloses in plant cell-walls.
- There are a number of procedures that are commonly used in many of the methods for dietary fiber analysis
 - Lipid removal:
 - Protein removal
 - Starch removal:
 - Selective precipitation of fibers:

Gravimetric Methods

- **Crude Fiber Method:** The crude fiber method gives an estimate of indigestible fiber in foods.
- It is determined by sequential extraction of a defatted sample with 1.25% H₂SO₄ and 1.25% NaOH.
- The insoluble residue is collected by filtration, dried, weighed and ashed to correct for mineral contamination of the fiber residue.
- Crude fiber measures cellulose and lignin in the sample
- but does not determine hemicelluloses, pectins and hydrocolloids, because they are digested by the alkali and acid and are therefore not collected

CH -4 EVALUATION OF DATA

- *Measures of Central Tendency*
 - Mean
 - Median
- *Reliability of Analysis*
- **Accuracy** refers to how close a particular measure is to the true or correct value
- **Precision:** This parameter is a measure of how reproducible or how close replicate measurements become.

Measure of Spread of Data

- The spread of the data is a measurement of how closely together repeated measurements are to each other.
- The standard deviation is the most commonly used measure of the spread of experimental measurements
 - SD = standard deviation
 - x_i = individual sample values
 - $\bar{x}$ = true mean
 - n = total population of sample

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

Sources of Error

- **Personal Errors (Blunders):** These occur when the analytical test is not carried out correctly: the wrong chemical reagent or equipment might have been used; some of the sample may have been spilt; a volume or mass may have been recorded incorrectly; etc

Random Errors:

- These produce data that vary in a non-reproducible fashion from one measurement to the next *e.g.*, instrumental noise
- **Systematic Errors:** A systematic error produces results that consistently deviate from the true answer in some systematic way, *e.g.*, measurements may always be 10% too high

Cont...

- Specificity
 - **Specificity** of a particular analytical method means that it detects only the component of interest
 - **Sensitivity** relates to the magnitude of change of a measuring device (instrument) with changes in compound concentration.
 - Sensitivity : the smallest concentration change that the method is capable of detecting

Rejecting Data

- There are several tests for rejecting an abnormal value.
- One of these tests, the **Q-Test**, is commonly used. In this test, a **Q-value** is calculated as shown below and compared to values in a table.
- If the calculated value is larger than the table value, then the questionable measurement can be rejected at the 90% confidence level
- It is one of the most statistically correct for a fairly small number of observations and is recommended when a test is necessary
- The ratio Q is calculated by arrange the data increasing order of numbers

Cont...

Table 3.3
Rejection Quotient, Q, at Different Confidence Limits*

No. of Observations	Confidence Level		
	Q ₉₀	Q ₉₅	Q ₉₉
3	0.941	0.970	0.994
4	0.765	0.829	0.926
5	0.642	0.710	0.821
6	0.560	0.625	0.740
7	0.507	0.568	0.680
8	0.468	0.526	0.634
9	0.437	0.493	0.598
10	0.412	0.466	0.568
15	0.338	0.384	0.475
20	0.300	0.342	0.425
25	0.277	0.317	0.393
30	0.260	0.298	0.372

*Adapted from D. B. Rorabacher, *Anal. Chem.*, **63** (1991) 139.

Cont...

- To apply a Q test for bad data, arrange the data in order of increasing values and Q is as defined:

$$Q = \frac{a}{w} \text{ or } Q = \frac{Q_{gap}}{Q_{range}}$$

Where,

- Q_{gap} or “a” is the absolute difference between the outlier in question (suspect number) and the closest number to it (nearest neighbor).
- Q_{range} or “w” is the difference between the highest number and the lowest number.
- If Q_{calculated} > Q_{table} then reject the questionable point.
- But, if Q_{calculated} < Q_{table} then the questionable point should not be rejected. ≥

- **EXAMPLE 1** For the data:
0.189, 0.169, 0.187, 0.183, 0.186, 0.182, 0.181, 0.184, 0.181, 0.177.
Determine if it can be ascribed to accidental error, at the 90% confidence level.

Solution:

Arranged it in increasing order:
0.169, 0.177, 0.181, 0.181, 0.182, 0.183, 0.184, 0.186, 0.187, 0.189

Thus, the outlier (suspected no.) is 0.169.

Then, calculate Q:

$$Q = \frac{a}{w} = \frac{(0.177 - 0.169)}{(0.189 - 0.169)} = 0.400$$

From the tabulated table, the 10 observations at 90% confidence level is 0.412.

Since, 0.4 (Q_{cal}) < 0.412 (Q_{table}). Therefore, 0.169 which is the outlier should not be rejected and can be ascribed to random error.

Example 2

A set of chloride analyses on separate aliquots of a pooled serum were reported:
103, 106, 107, and 130 meq/L.
One value appears suspect. Determine if it can be ascribed to accidental error, at the 95% confidence level.

The suspect result is 130meq/L.

$$Q = \frac{a}{w} = \frac{(130-107)meq / L}{(130-103)meq / L} = \frac{23}{27} = 0.852$$

From the tabulated table, the fourth observation at 95% confidence level is 0.829.

Since, 0.852 (Q_{cal}) > 0.829 (Q_{table}), the suspected number may be ascribed to random error and should be rejected

2/6/2022

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Cont...

Grubbs Test

The recommended way of identifying outliers is to use the Grubb's Test. A Grubb's test is similar to a Q-test however G_{exp} is based upon the mean and standard deviation of the distribution instead of the next-nearest neighbor and range.

$$G_{exp} = \frac{|x_q - \bar{x}|}{s}$$

Table: Critical Rejection Values for Identifying an Outlier: G-test

N	90% C.L.	95% C.L.	99% C.L.
3	1.153	1.154	1.155
4	1.463	1.481	1.496
5	1.671	1.715	1.764
6	1.822	1.887	1.973
7	1.938	2.020	2.139
8	2.032	2.127	2.274
9	2.110	2.215	2.387
10	2.176	2.290	2.482

If G_{exp} is greater than the critical G-value (G_{crit}) found in the Table then you are statistically justified in removing your suspected outlier.

Cont...

Grubbs Test for Outlier (Data Point)

Mass loss (%): 10.2, 10.8, 11.6, 9.9, 9.4, 7.8, 10.0, 9.2, 11.3, 9.5, 10.6, 11.6

Sidney Cheryl Yen Dick

Cheryl's value 7.8 looks out of line from the other data. A datum that is far from the other points is called an outlier. Should the group reject 7.8 before averaging the rest of the data or should 7.8 be retained?

We answer this question with the Grubbs test. First compute the average ($\bar{x}$) and the standard deviation (s) of the complete data set (all 12 points in this example):

$\bar{x} = 10.16 \quad s = 1.11$

Then compute the Grubbs statistic G , defined as

Grubbs test: $G = \frac{|\text{questionable value} - \bar{x}|}{s} \quad (4-9)$

If $G_{calculated} > G_{table}$, then the questionable value should be discarded!

$G_{calculated} = 2.13 \quad G_{table} \text{ (12 observations)} = 2.285$

Value of 7.8 should be retained in the data set.

Number of observations	G (95% confidence)
4	1.463
5	1.672
6	1.822
7	1.938
8	2.032
9	2.110
10	2.176
11	2.234
12	2.285
15	2.409
20	2.557

Calibration curve

- **calibration curve** is a general method for determining the concentration of a substance in an unknown sample by comparing the unknown to a set of standard samples of known concentration.
- The graphic or mathematic relationship between the readings obtained in an analytic process and the quantity of analyte in a calibration.
- The relationship is often a straight line rather than a curve.

Cont...

- The ability of a liquid to absorb, reflect, or transmit light varies with its concentration.
- These properties of a liquid can be measured in the laboratory with the help of an instrument called **spectrophotometer**.
- In order to construct a calibration curve, one needs a number of solutions of which the concentration is known, which are known as **standard solutions**.

How to Prepare Standard Solutions

- One can prepare standard solutions of the desired concentration in the laboratory by simply diluting/dissolving a carefully measured volume of either a stock solution (a concentrated laboratory reagent from which solutions can be made), or the substance itself in solid form, with a corresponding amount of water.
For example, to prepare a 10% solution of sodium chloride, 10 g of sodium chloride can be diluted in till the volume of solution become 100ml. Solutions of various concentrations can be prepared similarly.

Beer-Lambert's Equation

$$A = \epsilon lc$$

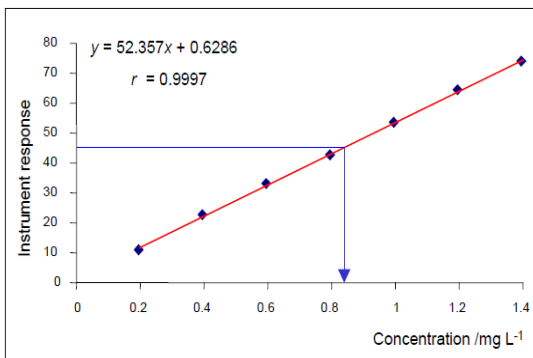
where,

$A \rightarrow$ absorbance of the solution

$\epsilon \rightarrow$ molar absorptivity (constant)

$l \rightarrow$ wavelength of light that passes through the solution

$c \rightarrow$ concentration of the solution



How to Draw a Calibration Curve

- The spectrophotometer is an analytical instrument that displays its output, or the absorbance of a liquid, in the form of deflections, caused due to the electrical signals generated by its inherent photodetector.
- The values observed can be mapped against the corresponding value of solution concentration on a graph paper.

Finding the Unknown Concentration

- The curve can be used to determine the concentration of an unknown solution.
- Using spectrophotometry, the absorbance of the unknown solution can be determined.
 - From the calibration curve, using graphical (interpolation from the graph)
 - mathematical methods (using Beer-Lambert's Equation), the corresponding value of solution concentration for the observed value of the unknown solution's absorbance can be obtained.

5. FOOD COMPOSITION DATA AND FOOD COMPOSITION DATABASES

- Food composition databases (FCDBs) or food composition tables are resources that provide detailed food composition data (FCD) on the nutritionally important components of foods.
- Early food composition studies were carried out to identify and determine the chemical nature of the principles in foods that affect human health

Methods of compiling food composition databases

1. Direct method

- The advantage of the direct method, in which all of the values are the results of analyses carried out specifically for the database being compiled, is that close control of the sampling, analysis and quality control procedures yields highly reliable data.

2. Indirect method

- The indirect method uses data taken from published literature or unpublished laboratory reports.
- There is consequently less control over the quality of the data, which may be uneven. Great care must therefore be taken in their appraisal/assessment/ for inclusion in the database.

3. Combination method

- Most food composition databases nowadays are prepared by a combination of the direct and indirect methods, containing original analytical values together with values taken from the literature and from other databases as well as imputed and calculated values
- Four issues about food composition data are particularly important:
 - (a) the representativeness of the data;
 - (b) the sources of data;
 - (c) the method used to derive nutrient values
 - (d) the adequacy/capability/ of analytical methodologies

Cont...

- The representativeness of data depends on the number and quality of laboratories performing food composition analyses, the number and quality of samples used, and any weighting procedures used.
- Representative data reflect the nutrient composition of food products on a nationwide, year-round basis

Cont...

- In general order of preference, the sources of data are:
 - **Original analytical values:** These are values taken from the published literature or unpublished laboratory reports, whether or not they were from analyses carried out explicitly for the purpose of compiling the database.
 - **Imputed values:** These data are estimates derived from analytical values obtained for a similar food (e.g. values for peas used for green beans) or for another form of the same food (e.g. values for “boiled” “used for “steamed”).
 - **Calculated values:** These are values derived from recipes, calculated from the nutrient contents of the ingredients and corrected for preparation factors: loss or gain in weight, usually referred to as yields, and micronutrient changes, usually referred to as retention factors
 - **Borrowed values:** These are values taken from other tables and databases where reference back to the original source may or may not be possible
 - Adequate reference to the original source is necessary to justify since there may be d/t water or fat contents of food
 - **Presumed values:** These are values presumed as being at a certain level or as zero, according to regulations

Sources of food composition data

- The main source of food composition data was the United States Department of Agriculture (USDA).
The rationale for this selection was:
 - It is a source that is updated at regular intervals.
 - It is reliable in terms of quality. Food data admitted to this FCT are obtained using up-to-date analytical methods by the USDA or contracted partners.
 - Nutrient data for both raw and one or more cooked forms of the same food are often available

Types of FCDB

1. Simplified food composition database or tables

- Simplified databases or tables can be produced from the main user database.
- In these, fewer nutrients are covered, and some reductions of food categories may be possible (e.g. for meat cuts data may appear only for “medium cooked,” omitting “raw” and “well cooked”).

Cont...

- **Special-purpose food composition tables and databases**
 - Tables and databases restricted to selected nutrients can be produced for people with special dietary needs or interests (e.g. for diabetics, or for people with kidney disorders for whom a diet controlling protein, sodium and potassium is required, or for nutrition educators, or for people wishing to lose weight).
 - Data may be presented per 100 g of food, or per portion size or common household measures.

Cont...

- *Types of food composition database programme*
- 1. National
- 2. Regional

Criteria for a comprehensive FCDB

- The current high level of interest in nutrition requires that food composition databases meet the following criteria:
 - **Data should be representative**
 - **Data should be of sound analytical quality**
 - **Coverage of foods should be comprehensive**
 - The database should include all foods that form a major part of the food supply and as many as possible of the less frequently consumed foods.
 - **Coverage of nutrients should be comprehensive**
 - Values should be included for all of the nutrients and other components known or believed to be important to human health

Cont...

- Food descriptions should be clear
- Data should be consistently and unambiguously expressed
- Origins of data should be provided at nutrient value level
- Tables and databases should be easy to use
- Database should have few missing data

Uses of food composition data

1. Food composition data are used primarily for the assessment and the planning of human energy and nutrient intakes
- Assessment of nutrient intakes (nutritional analysis)
 - Individual level
 - Group level
 - National and international levels
 - Sub-national and community levels

Cont...

2. Prescription of therapeutic diets:

- A therapeutic diet must be nutritionally balanced and adequate while at the same time controlling the intake of one or more specified nutrients

3. Planning of institutional diets:

- Food composition data are used to translate recommended nutrient intakes into cost-limited foods and menus

4. National food and nutrition policy:

- A national food and nutrition policy will often define goals for the intake of certain nutrients

Cont....

5. Nutritional regulation of the food supply:

- Food regulators use nutritional data on primary foods or “traditional” food products as a reference point for desirable nutrient levels for processed and newly introduced foods

6. Planning of nutrition intervention programmes:

- Nutrition interventions, such as food aid programmes, supplementation schemes and disease prevention programmes, require the use of food composition data in order to translate specific nutrient needs into food requirements

Limitations of food composition databases

- The limitations of food composition tables or databases are often not sufficiently understood by many users.
- But its limitation is Foods, being biological materials, exhibit variations in composition; therefore a database cannot accurately predict the composition of any given single sample of a food.

FAO can assist the regional food composition network in the following ways:

- 1. Formulate up-to-date standards and procedures that specify minimum quality criteria required for food composition data
- 2. Assist in selection of priority constituents of food to be analyzed .such as nutrient additives, natural toxins and contaminants in foods. These priorities should be associated with public health problems and trade in the region
- 3. Assists in the introduction of quality assurance systems so that the quality of the food composition data is accurate and valid.

Cont...

4. Provide information and guidance on sampling procedures selection of analytical methodologies and above all training manuals
5. Play major role in the dissemination of and continuous revision of data on food composition that occur due to changes in product formulation and food processing techniques ,food varieties ,agriculture production systems and improvements in analytical techniques.